

1.11 Differential Equations

Mathematics B (MEI) — OCR B — A-Level (H640) — Pure Mathematics — spec refs c36–c39 (Calculus)

What this topic covers. Forming differential equations from a described rate of change, solving them by separating the variables, applying initial conditions, and interpreting and criticising the resulting models.

1. What a differential equation is

A **differential equation** relates a function to its derivatives. A **first order** equation involves only the first derivative.

Solving one produces a **general solution** containing an arbitrary constant. Applying an initial or boundary condition pins that constant down and gives the **particular solution**.

2. Forming a differential equation

Translate the wording of the rate directly.

Statement	Equation
The rate of growth is proportional to the amount present	$\frac{dP}{dt} = kP$
The rate of decay is proportional to the amount present	$\frac{dN}{dt} = -kN$
The rate is proportional to the square of the population	$\frac{dP}{dt} = kP^2$

The rate of cooling is proportional to the excess temperature	$\frac{d\theta}{dt} = -k(\theta - \theta_0)$
The quantity approaches a limit A at a rate proportional to the shortfall	$\frac{dy}{dt} = k(A - y)$

"Proportional to" introduces a constant k ; a *decreasing* quantity takes a minus sign, with k itself kept positive. Writing $\frac{dN}{dt} = kN$ for decay and then finding k negative is acceptable but muddles later interpretation.

3. Separating the variables

If the equation can be written as $\frac{dy}{dx} = f(x)g(y)$, gather each variable on its own side and integrate:

$$\int \frac{1}{g(y)} dy = \int f(x) dx$$

Worked example. Solve $\frac{dy}{dx} = xy$, given $y = 1$ when $x = 0$.

$$\int \frac{1}{y} dy = \int x dx \implies \ln y = \frac{x^2}{2} + c$$

At $x = 0$, $y = 1$: $\ln 1 = 0$, so $c = 0$. Hence

$$y = e^{x^2/2}$$

Worked example. Solve $\frac{dy}{dx} = \frac{y}{x}$ for $x > 0$.

$$\int \frac{1}{y} dy = \int \frac{1}{x} dx \implies \ln y = \ln x + c$$

Writing $c = \ln A$ gives $\ln y = \ln(Ax)$, so $y = Ax$.

Replacing the constant by $\ln A$ turns an additive constant into a multiplicative one and keeps the final answer tidy. Similarly, $y = e^{x+c}$ is better written $y = Ae^x$ with $A = e^c$.

Worked example. Solve $\frac{dy}{dx} = 2xy^2$.

$$\int y^{-2} dy = \int 2x dx \implies -\frac{1}{y} = x^2 + c$$

Worked example. Solve $\frac{dy}{dx} = y \tan x$.

$$\int \frac{1}{y} dy = \int \tan x dx \implies \ln y = \ln |\sec x| + c \implies y = A \sec x$$

4. The exponential model

The single most important case:

$$\frac{dP}{dt} = kP \implies P = P_0 e^{kt}$$

with P_0 the value at $t = 0$. Decay is the same with $-k$.

Worked example. Solve $\frac{dy}{dt} = 3y$ with $y = 5$ at $t = 0$.

$y = Ae^{3t}$, and $y(0) = 5$ gives $A = 5$, so $y = 5e^{3t}$.

5. Newton's law of cooling

Worked example. A body cools according to $\frac{d\theta}{dt} = -k(\theta - 20)$, with $\theta = 100$ at $t = 0$. Solve, and state the long-term temperature.

$$\int \frac{1}{\theta - 20} d\theta = \int -k dt \implies \ln(\theta - 20) = -kt + c$$

$$\theta - 20 = Ae^{-kt}$$

At $t = 0$, $\theta = 100$, so $A = 80$:

$$\theta = 20 + 80e^{-kt}$$

As $t \rightarrow \infty$ the exponential tends to zero, so $\theta \rightarrow 20^\circ\text{C}$ — the surrounding temperature, as physical sense demands.

Checking the long-term behaviour is a free sanity check on your answer. A cooling model that tends to anything other than the ambient temperature contains an algebraic error.

6. Approach to a limit

The equation $\frac{dy}{dt} = k(A - y)$ describes a quantity rising towards a ceiling A , with the rate falling as the gap closes. Its solution has the form $y = A - Be^{-kt}$, which approaches but never reaches A .

The same structure appears in the mechanics of a falling body with resistance, $\frac{dv}{dt} = g - kv$, whose limit $v = \frac{g}{k}$ is the **terminal velocity** — the value at which the acceleration becomes zero.

7. The logistic model

Unlimited exponential growth is unrealistic for a population with finite resources. The logistic equation builds in a ceiling:

$$\frac{dy}{dt} = ky(1 - y)$$

Separating requires **partial fractions**, since

$$\frac{1}{y(1 - y)} = \frac{1}{y} + \frac{1}{1 - y}$$

Recognising that a logistic-type equation needs partial fractions before it can be integrated is the step that unlocks the whole question. If separation leaves you with a product in the denominator, split it first.

8. Criticising a model

MEI asks for evaluation, not just solution. Points that reliably earn credit:

- Exponential growth predicts unbounded increase, ignoring limited food, space or resources.
- A decay model tending to zero may be unrealistic if a residual background level remains.
- The constant k is assumed fixed, but may drift with temperature, season or other conditions.
- Populations are discrete, whereas the model treats them as continuous.
- Extrapolating far beyond the range of the data that fitted the model is not justified.

9. Common pitfalls

- Forgetting the constant of integration, so no initial condition can be applied.
- Applying the initial condition before integrating.
- Leaving the answer as $\ln y = \dots$ when the question asks for y .
- Sign errors on a decay constant, giving growth instead of decay.
- Attempting to separate an equation that is not separable.
- Missing that a product in the denominator needs partial fractions.
- Not stating the long-term behaviour when the question asks for it.